- **Exercise:**

1. If your client needs three groups, how will you create a groups, how many groups will you create? find the IP range, Host range, network id, broadcast id, of each group. **(Consider IP 200.1.2.0).**

1. Group creation Strategy
Number of groups needed: 3
Bits to borrow: To create at least 3 groups, we use the formulla $2^n \geq 3$, where n is the number
$2^2 = 4$ subnets

2. New Subnet Mask
Default class (mask) = 255.255.255.0 (/24)
After barrowing 2 bits: 126.255.255.255.192

3. Block Size
256 - 192 = 64

Broadcast ID = 200.1.2.127
IP Range: 200.1.2.65 - 200.1.2.126
Broadcast ID = 200.1.2.65
IP Range: 200.1.2.1 to 200.1.2.6.62

Brodcast ID = 200.1.2.191.
IP Range: 200.1.2.199
Broadcast ID = 200.1.2.255
IP Range: 200.1.2.193

2. In a class B, network on the internet has a subnet mask 255.255.240.0 what is the max. nr. of hosts per subnet.

Network: 200.1.2.0
Subnet Mask: 255.255.255.240

1. find Prefix
255.255.255.240 = /28

2. Total IPs in Subnet
$2^{(32-28)} = 2^4 = 16$

3. Usable Hosts
16 - 2 = 14

4. First Subnet Block
256 - 240 = 16

Range:
Network Address: 200.12.0
First Hosts : 200.1.2.1
last Host : 200.1.2.14
Broadcast Address : 200.1.2.15

3. Given IP Address – 172.16.0.0/25, find the number of subnets and the number of hosts per subnet. Also, for the first subnet block, find the subnet address, first host ID, last host ID and broadcast address.

1. Default Class
IP 172.16.0.0 is class B
Default mask = 255.255.0.0(/16)
Give mask = /25
Broadcast bits:
25-16 = 9

2. Number of Subnets
$2^9 = 512$
Number of subnet = 512

3. Hosts Per subnet
Hosts mask /25 = 255.255.255.128
= 32-25 = 7
$= 2^7 = 2 = 126$

4. First Subnet Block
Subnet mask /25 = 255.255.255.128
Block size = 128
First subnet:
subnet Address : 172.16.0.0 Last Host : 172.16.0.126
First hots : 172.16.0.01 Broadcast Address : 172.16.0.127

4. In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____

(A) 158 (B) 255 (C) 222 (D) 223

5. In the IPv4 addressing format, the number of networks allowed under Class C addresses is

(A) 2^{14} (B) 2^{7} (C) 2^{21} (D) 2^{24}

6. Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively and they both use the same net mask N. Which of the values of N given below should not be used if A and B should belong to the same network? _____

(A) 255.255.255.0 (B) 255.255.255.128 (C) 255.255.255.192 (D) 255.255.255.224

7. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(A) 1022 (B) 1023 (C) 2046 (D) 2047

8. What is the broadcast address of the network *172.31.180.131/25*?
OR What is the broadcast address of the network *172.31.180.128/25*?

Given IP 172.31.180.131/25

1. Subnet mask
Prefix 125 = 255.255.128
This divides the last octent into blocks of 128

0-127
128-255
Identify the Subnet
IP 172.31.180.131 falls in the range 128-125

Network Address: 172.31.180.128
Blockcast Address: 172.31.180.255

1. Subnet mask
125 = 255.255.255.128

2. Address Range
Block size = 128

subnet range:
Network Address = 172.31.180.1
Broadcast Address = 172.31.180.2

9. What valid host range is the IP address 10.254.201.56/20 a part of?

1. Subnet Mask Prefix /20 means the subnet mask is 255.255.240.0
Block size in the 3rd octent 256-240 = 16
Subnet in the 3rd octet: 0, 16, 32, 46, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 244, 240

2. Idenify the Subnet
The IP is 10.254.20.156
201 falls between 192 and 207, so the subnet is.
Network Address: 10.254.192.0
Broadcast Address: 10.254.207.255

3. Valid Host Range:
First Host: 10.254.192.1
Last Host: 10.254.207.254

10. How many subnets and hosts per subnet can you get from the network *172.28.0.0/23?*

1. Default Network Class

→ The IP 172.28.0.0 belongs to Class B
But the given prefix is /23, so additional bits are used for subnetting.

2. Number of Host Bits

Total bits = 32
32 - 23 = 9 hots bits
Number of hosts per subnet:
$2^9 - 2 = 512 - 2 = 510$
Hosts per subnet = 510

11. What is the last valid host on the subnetwork *172.19.156.0/23?*

→ 1. Subnet Information

Network Address: 172.192.156.0
Prefix : /23
Subnet mask : 255.255.254.0

The /23 subnet means 9 bits are available usable hosts range:
for hosts

Total IPs = 2^9 = 512
1 Network Address
1 Broadcast Address

2. Find the Range

Subnet range:
Network Address: 172.19.156.0
Broadcast Address: 172.19.157.255
First Hosts : 172.19.19.156
Last Hosts: 172.19.19.257

→ Broadcast the last valid hosts in subnet 172.19.156.0/23 is 172.19.157.254

→ Broadcast - 1 = last host
172.19.157.255 - 1 = 172.19.17.154

12. What valid host range is the IP address 192.168.206.28 and subnet mask is 255.255.255.248 a part of?

step:1
Convert the Subnet mask 255.255.255.248 to binary:
- 255 - 11111111
- 255 - 11111111
- 255 - 11111111
- 248 - 11111000 so the subnet mask in binary is 11111111.11111111.11111111.11111000

step:2
Define the block size in the last octent:
Block size = 256 - 248 = 8

step:3 Find the subnet that contains the IP address 192.168.206.28:
- The subnet in the last octent will start at multiple of the : 8, 16, 24, 32
- Since 28 is between 24 and 31 the subnet range is 192.168.206.24 to 192.168.206.31

Step:4
Identify the network address and broadcast address:
- Network Address: 192.168.206.24 (first address in the subnet)
- Broadcast Address: 192.168.206.25 (last address)

Conclusion:
It is conduded that subneting is an esstinal the technique in computer networking that allow a large network to be divided into smaller, efficient and manageable subnetwork. By borrowing bits from the host portion of an IP address multiple subnets can be created within a single class A, B, C Network.

Signature with Date of Completion	
Marks out of 10	

Practicle 10

1. What is scope?
In DHCP terminology, a "Scope" refers to the specific range of consecutive IP addresses that are available on a DHCP server for leasing to clients on a particular subnet. A scope defines the "Pool" of addresses (e.g., 192.168.1.10 to 192.168.1.100). Besides just the IP addresses, a scope also includes critical configuration parameters such as the Subnet Mask, the Default Gateway, the DNS Server addresses, and the Lease Duration (how long a client can keep the IP).

2. List types of IP address Allocation methods with its brief.

Dynamic Allocation: The most common method where the server "leases" an IP from the pool for a set time. When the lease expires, the IP returns to the pool.

Automatic Allocation: Similar to dynamic, but the server keeps a table of past assignments so that the client is permanently assigned the same IP every time it connects.

Static/Manual Allocation: The administrator manually maps a client's MAC address to a specific IP address in the server configuration. The device will always receive that exact same IP, which is essential for network printers and servers.

3. What is IP reservation?
IP Reservation is a feature of DHCP that ensures a specific device on the network always receives the same IP address every time it requests one. This is achieved by creating a permanent mapping between the device's unique MAC (hardware) address and a specific IP address within the DHCP scope. Unlike a manual "Static IP" configured on the device itself, a reservation is managed centrally on the DHCP server, making it easier to track and manage from a single location.

Conclusion:
This practical successfully demonstrated the automation of network configuration using the Dynamic Host Configuration Protocol (DHCP). We implemented the DORA process (Discover, Offer, Request, Acknowledgment) through simulation. By configuring a DHCP pool on a router, we observed how client PCs automatically obtained their IP addresses and gateway settings without any manual intervention. We concluded that DHCP is essential for reducing administrative overhead, preventing IP address conflicts (duplicate IPs), and ensuring that mobile devices can move between network segments seamlessly while maintaining connectivity.